1. Title Page

Lab Name: Lab 08: Router Architecture & Forwarding **Date:** YYYY-MM-DD **Name:** [Your Name] **Lab Partners:** [Partners' Names, if any] **Software/Hardware Used:** Cisco Modeling Labs (CML) - Personal / Free Edition

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2. Background

Routers are Network Layer devices responsible for forwarding packets end-to-end between source and destination hosts. A router's internal architecture is logically split into two planes:

1. **Control Plane (Software-based)**: Computes routing tables (RIB), runs routing protocols (e.g. OSPF, BGP), and handles management tasks. Operates in the millisecond timeframe.
2. **Data Plane (Hardware-based)**: Physically forwards packets from an input interface to the appropriate output interface using a Forwarding Information Base (FIB) and Adjacency Table. Operates in the nanosecond timeframe.

Inside the data plane, routers perform decentralizing lookup (matching destination prefixes in TCAM tables), switch packets across a high-speed switching fabric (memory, bus, or crossbar), queue packets in buffers during output-link congestion, and schedule departures (e.g., FCFS, Priority, Weighted Fair Queueing).

In this lab, you will configure a triangle topology of Cisco IOSv routers, verify static routing, and query the router data-plane tables (CEF and Adjacency) to see how theoretical router architectures are implemented in real networking devices.

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3. Objective / Purpose

To configure static routing on a multi-router network, analyze the difference between the routing table (RIB) and the forwarding table (FIB), inspect Layer 2 rewrite adjacency tables (Cisco Express Forwarding), and map theoretical router architecture components to CLI commands and packet captures.

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4. Topology Diagram

```
graph TD
    R0["Router R0<br>Gi0/0: 10.1.12.1/24<br>Gi0/1: 10.1.13.1/24"] <--> |"10.1.12.0/24"| R1["Router R1<br>Gi0/0: 10.1.12.2/24<br>Gi0/1: 10.1.23.2/24"]
    R0 <--> |"10.1.13.0/24"| R2["Router R2<br>Gi0/0: 10.1.13.3/24<br>Gi0/1: 10.1.23.3/24"]
    R1 <--> |"10.1.23.0/24"| R2

    classDef router fill:#fff2cc,stroke:#d6b656,stroke-width:2px;
    class R0,R1,R2 router;
```

Details:

- 3x Routers in a triangle topology.
- Node names: `<initials>-R0`, `<initials>-R1`, `<initials>-R2`.

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5. Equipment & Environment

- **Software:** Cisco Modeling Labs (CML), Terminal Emulator (e.g., PuTTY, SecureCRT, native terminal), Wireshark
- **Hardware / Virtual Nodes:** 3x Cisco IOSv Routers

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6. Configuration / Methodology

IP Addressing Table

Device	CML Node Name	Interface	IP Address	Subnet Mask
R0	`<initials>-R0`	G0/0	10.1.12.1	255.255.255.0
R0	`<initials>-R0`	G0/1	10.1.13.1	255.255.255.0
R1	`<initials>-R1`	G0/0	10.1.12.2	255.255.255.0
R1	`<initials>-R1`	G0/1	10.1.23.2	255.255.255.0
R2	`<initials>-R2`	G0/0	10.1.13.3	255.255.255.0
R2	`<initials>-R2`	G0/1	10.1.23.3	255.255.255.0

Step-by-Step Configuration Guide

1. Configure Router R0 (`<initials>-R0`)

Open R0's console, elevate privileges, and configure IP addressing and static routing:

```
Router> enable
Router# configure terminal
Router(config)# hostname KH-R0
KH-R0(config)# interface GigabitEthernet0/0
KH-R0(config-if)# ip address 10.1.12.1 255.255.255.0
KH-R0(config-if)# no shutdown
KH-R0(config-if)# exit
KH-R0(config)# interface GigabitEthernet0/1
KH-R0(config-if)# ip address 10.1.13.1 255.255.255.0
KH-R0(config-if)# no shutdown
KH-R0(config-if)# exit
KH-R0(config)# ip route 10.1.23.0 255.255.255.0 10.1.12.2
KH-R0(config)# end
KH-R0# write memory
```

(The static route allows R0 to reach the subnet between R1 and R2 by sending packets to R1's IP `10.1.12.2` *).*

2. Configure Router R1 (`<initials>-R1`)

```
Router> enable
Router# configure terminal
Router(config)# hostname KH-R1
```

```
KH-R1(config)# interface GigabitEthernet0/0
KH-R1(config-if)# ip address 10.1.12.2 255.255.255.0
KH-R1(config-if)# no shutdown
KH-R1(config-if)# exit
KH-R1(config)# interface GigabitEthernet0/1
KH-R1(config-if)# ip address 10.1.23.2 255.255.255.0
KH-R1(config-if)# no shutdown
KH-R1(config-if)# exit
KH-R1(config)# ip route 10.1.13.0 255.255.255.0 10.1.12.1
KH-R1(config)# end
KH-R1# write memory
```

3. Configure Router R2 (`<initials>-R2`)

```
Router> enable
Router# configure terminal
Router(config)# hostname KH-R2
KH-R2(config)# interface GigabitEthernet0/0
KH-R2(config-if)# ip address 10.1.13.3 255.255.255.0
KH-R2(config-if)# no shutdown
KH-R2(config-if)# exit
KH-R2(config)# interface GigabitEthernet0/1
KH-R2(config-if)# ip address 10.1.23.3 255.255.255.0
KH-R2(config-if)# no shutdown
KH-R2(config-if)# exit
KH-R2(config)# ip route 10.1.12.0 255.255.255.0 10.1.13.1
KH-R2(config)# end
KH-R2# write memory
```

Step-by-Step Traffic Generation & Execution:

1. **Verify Connectivity**: Open `<initials>-R0`'s console and ping R2's interface on the opposite link:

   ```
   KH-R0# ping 10.1.23.3
   ```

   (Ensure you get a success rate of 100 percent, verifying that your static routes are correct).
2. **Start Packet Capture**: Right-click the link between **Router R0 and Router R1**, select **Packet Capture**, and click **Start**.
3. **Trigger ARP Resolution**: Open `<initials>-R0`'s console and clear the ARP cache to force a new lookup:

   ```
   KH-R0# clear arp
   KH-R0# ping 10.1.12.2
   ```

4. **Collect CLI Outputs**: Run the following verification commands on `<initials>-R0` and record the results:
   - `show ip route` (Displays the control plane routing table).
   - `show ip cef` (Displays the data plane forwarding table).
   - `show adjacency detail` (Displays the L2 rewrite adjacency table).
   - `show arp` (Displays the resolved IP-to-MAC associations).

5. **Download Capture**: Stop the CML packet capture and download the `.pcap` capture file to your local computer.

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7. Verification and Analysis Questions

Analyze the CLI outputs collected from `<initials>-R0` and the packet capture in Wireshark, and answer the following questions based on the concepts in the Chapter 4 lecture slides:

1. Control Plane vs. Data Plane CLI Mapping

- Examine the outputs of `show ip route` and `show ip cef` on R0.
- Contrast the two outputs. Which command displays the RIB (Routing Information Base) belonging to the **control plane**, and which displays the FIB (Forwarding Information Base) belonging to the **data plane**?
- Based on the slides, explain the difference in operation speed (millisecond vs. nanosecond timeframe) and hardware implementation between these two planes.

2. Longest Prefix Matching

- In destination-based forwarding, how does a router determine which table entry to use when a packet destination IP matches multiple route entries?
- Does Cisco IOS use longest prefix matching? Illustrate how the IP prefix `10.1.12.0/24` in your routing table corresponds to a range of addresses, and explain what happens if a packet arrives for `10.1.12.2`.

3. CEF Adjacency and Layer 2 Rewrite (Match-plus-Action)

- Inspect the output of the `show adjacency detail` command on `<initials>-R0`.
- Look at the hexadecimal output representing the Layer 2 rewrite string (adjacencies). Explain how this maps to the "link layer encapsulation" function performed in a router's **output port** interface card before sending bits onto the physical link.
- In what way does this represent a "Match-plus-Action" model?

4. Router Switching Fabric Architecture

- What are the three primary types of **switching fabrics** used inside routers as detailed in Slide 25?
- Cisco IOSv is a software-based virtual router running inside a hypervisor on x86 server hardware. Which switching fabric architecture does it functionally simulate/rely on for transferring packets between virtual interfaces? What is the main bottleneck of this fabric type?

5. Queuing and HOL Blocking

- Define **Head-of-the-Line (HOL) blocking** and explain under what circumstances it occurs (does it happen at the input ports or output ports of a switching fabric?).
- If a router's switching fabric is slower than the combined input ports, where will queuing occur? If the fabric is faster, where will queuing occur?

6. Buffer Management and Sizing

- According to Slide 33 (RFC 3439 rule of thumb), how much buffer capacity is recommended for a router interface operating at speed C with a round-trip time RTT?
- If one of `<initials>-R0`'s interfaces was a physical 10 Gbps port with a typical network RTT of 120 ms, calculate the required buffer size in Megabits.
- Explain the negative consequence of having too much buffering on a link (bufferbloat).

7. Scheduling and Net Neutrality

- Briefly define the **Weighted Fair Queueing (WFQ)** scheduling policy. How does it guarantee bandwidth to specific traffic classes compared to simple FCFS (FIFO)?
- Explain how packet scheduling and buffer management are technical mechanisms directly related to the social, political, and economic arguments surrounding **Network Neutrality** (specifically the rules against throttling and paid prioritization).

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8. Troubleshooting

Issue Encountered: [Describe what went wrong, if anything] **Discovery Method:** [How did you find the issue?] **Resolution:** [How did you fix it?]

(If no issues were encountered, explain potential pitfalls for this configuration).

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9. Conclusion & Analysis

Summarize what you learned about router architectures. Explain the roles of lookup tables, switching fabrics, queuing, and scheduling policies in enabling high-performance packet forwarding.

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10. What to Hand In

- The Lab YAML configuration file with your name, date, and name of the lab at the top in comments.
- A single PDF report containing:
  - Your written answers to the 7 analysis questions in Section 7.
  - A screenshot of your active CML topology showing the running routers (`<initials>-R0` , `<initials>-R1` , `<initials>-R2`).
  - A screenshot of the routing table (`show ip route` command output) for **each** of the three routers (`<initials>-R0` , `<initials>-R1` , and `<initials>-R2`).
  - The required screenshot of the router CLI console displaying `<initials>-R0` 's `show ip interface brief` status.
  - An annotated screenshot of your Wireshark capture window showing the ARP Request and ARP Reply exchange between R0 and R1.